

Chapter 8 Confidence Intervals

We have seen that the **sample mean is the point estimate** of the **population mean** and the **sample proportion** is the point estimate of the **population proportion**. In general, although a point estimate is a reasonable one number estimate of a population parameter (mean, proportion, or the like), the point estimate will not—unless we are extremely lucky—equal the true value of the population parameter. In this chapter we study **how to use a confidence interval to estimate a population parameter**. A **confidence interval** for a population parameter is an interval, or range of numbers, constructed around the point estimate so that we are very sure, or confident, that the true value of the population parameter is inside the interval.

By computing such an interval, we estimate—with confidence—the possible values that a population parameter might equal. This, in turn, can help us to assess—with confidence—whether a particular business improvement has been made or is needed.

Outline

- Section 8.1 z -Based Confidence Intervals for a Population Mean: σ known.
- Section 8.2 t -Based Confidence Intervals for a Population Mean: σ unknown.
- Section 8.3 Sample Size Determination.
- Section 8.4 Confidence Intervals for a Population Proportion.

Section 8.1

z-Based Confidence Intervals for a Population Mean: σ known

- The starting point is the **sampling distribution of the sample mean**:
 - Recall from Chapter 7 that if a population is normally distributed with mean μ and standard deviation σ , then the sampling distribution of $\bar{x}$ is normal with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. (**Note:** for $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ holds, the populations are at least 20 times the size of the sample)
 - Use a normal curve as a model of the sampling distribution of the sample mean
 - **Exactly**, because the population is normal.
 - **Approximately**, by the Central Limit Theorem for large samples.

- Recall the empirical rule, so...
 - 68.26% of all possible sample means are within one standard deviation of the population mean.
 - 95.44% of all possible sample means are within two standard deviations of the population mean.
 - 99.73% of all possible sample means are within three standard deviations of the population mean.

- A confidence interval for a population mean is an interval constructed around the sample mean so that we are reasonably sure, or confident, that this interval contains the population mean.
- Any confidence interval for a population mean is based on what is called a confidence level. This confidence level is a percentage (for example, 95 percent or 99 percent) that expresses how confident we are that the confidence interval contains the population mean.

Example 8.1 The Car Mileage Case

- Before the automaker selected the sample of $n = 50$ new midsize cars and tested them, there were many samples of 50 cars and corresponding mileages that the automaker might have obtained. Because different samples generally have different sample means, we consider the probability distribution of the population of all possible sample means that would be obtained from all possible samples of $n = 50$ car mileages.
- In Chapter 7 we have seen that such a probability distribution is called the sampling distribution of the sample mean. Several of these properties tell us that, if the population of all individual midsize car mileages is normally distributed with mean μ and standard deviation σ , then for any sample size n the sampling distribution of the sample mean is a normal distribution with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n}$.

- Recall that the population of car mileages is normally distributed with mean μ and standard deviation $\sigma = 0.8$ mpg
 - Note that μ is unknown and is to be estimated.
- Taking samples of size $n = 50$, the sampling distribution of sample mean mileages $\bar{x}$ is normal with mean $\mu_{\bar{x}} = \mu$ (which is also unknown) and standard deviation

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{50}} = 0.113$$

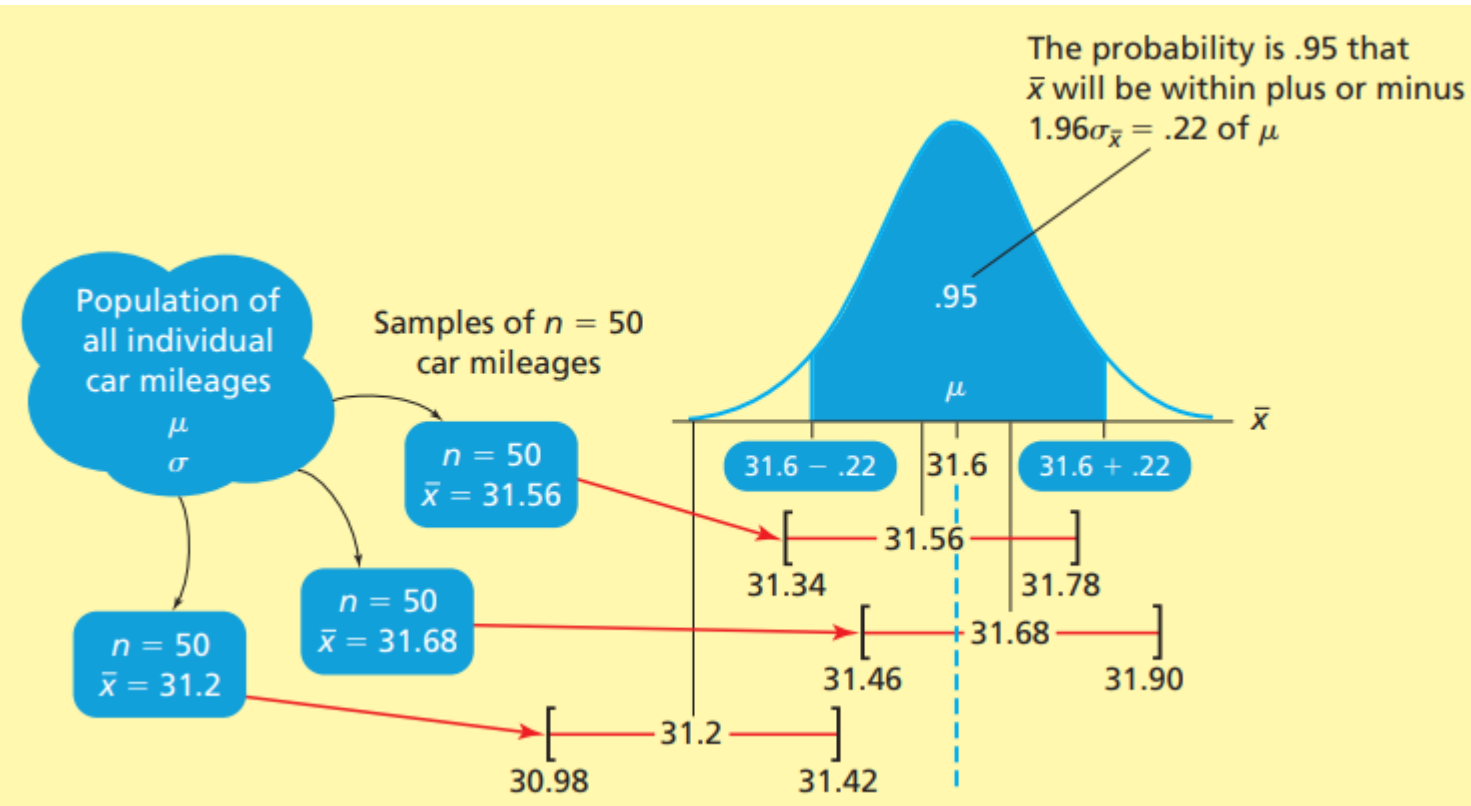
- The probability is 0.95 that $\bar{x}$ will be within plus or minus $1.96\sigma_{\bar{x}} = 1.96 \cdot 0.113 = 0.22$ of μ . (we know this in Chapter 6)

- The sample mean is within ± 0.22 of μ

$$P(\mu - 0.22 \leq \bar{x} \leq \mu + 0.22) = 0.95$$

- Then there is a 0.95 probability that $\bar{x}$ will be a value so that interval $[\bar{x} \pm 0.22]$ contains μ .
 - The interval $[\bar{x} \pm 0.22]$ is referred to as the 95% **confidence interval** for μ .
 - 0.22 is called the **margin of error** when estimating μ by $\bar{x}$.

Figure 8.2 Three 95% Confidence Intervals for μ



- Three intervals shown, two contain μ , one does not.

Assumes that the true value of the population mean μ is 31.6. Then, because the sample mean $\bar{x} = 31.56$ is within 0.22 of $\mu = 31.6$, the interval $[31.34, 31.78]$ contains μ , and the corresponding probability is 0.95.

However, because the sample mean $\bar{x} = 31.2$ is not within 0.22 of $\mu = 31.6$, the interval $[31.2 \pm 0.22] = [30.98, 31.42]$ does not contain μ .

- According to the 95% confidence interval, we know...
 - There is 95% probability that the sample mean that is calculated is such that the interval $[\bar{x} \pm 0.22]$ will contain the actual (but unknown) population mean μ .
 - There is a 5% probability that the interval does not contain μ .
- In other words, of all possible sample means, 95% of all the corresponding intervals $[\bar{x} \pm 0.22]$ will contain the population mean μ .

For example, recall that when the automaker randomly selected the sample of $n = 50$ cars and tested them, the automaker obtained the sample of 50 mileages given in Table 1.6. The mean of this sample is $\bar{x} = 31.56$ mpg, and a histogram constructed using this sample (see Figure 2.9) indicates that the population of all individual car mileages is normally distributed. It follows that a 95 percent confidence interval for the population mean mileage μ of the new midsize model is $[\bar{x} \pm 0.22] = [31.56 \pm 0.22] = [31.34, 31.78]$.

Because we do not know the true value of μ , we do not know for sure whether this interval contains μ . However, we are 95 percent confident that this interval contains μ . That is, we are 95 percent confident that μ is between 31.34 mpg and 31.78 mpg.

Generalizing

- In the example, we found the probability that μ is contained in an interval (confidence interval is given, to find the probability).
- More usual to specify the probability and find the corresponding interval (probability is given, to find the confidence interval).
- The probability that the confidence interval will **not** contain the population mean μ is denoted by α
 - In the above example, $\alpha = 0.05$.

Generalizing - continued

- The probability that the confidence interval will contain the population mean μ is denoted by $1 - \alpha$
 - $1 - \alpha$ is referred to as the confidence coefficient
 - $(1 - \alpha) \times 100\%$ is called the **confidence level**
 - In the example, the confidence coefficient is 0.95, and the confidence level is 95%.
- Usual to use two decimal point probabilities for $1 - \alpha$
 - Here, focus on $1 - \alpha = 0.95$ or 0.99 .

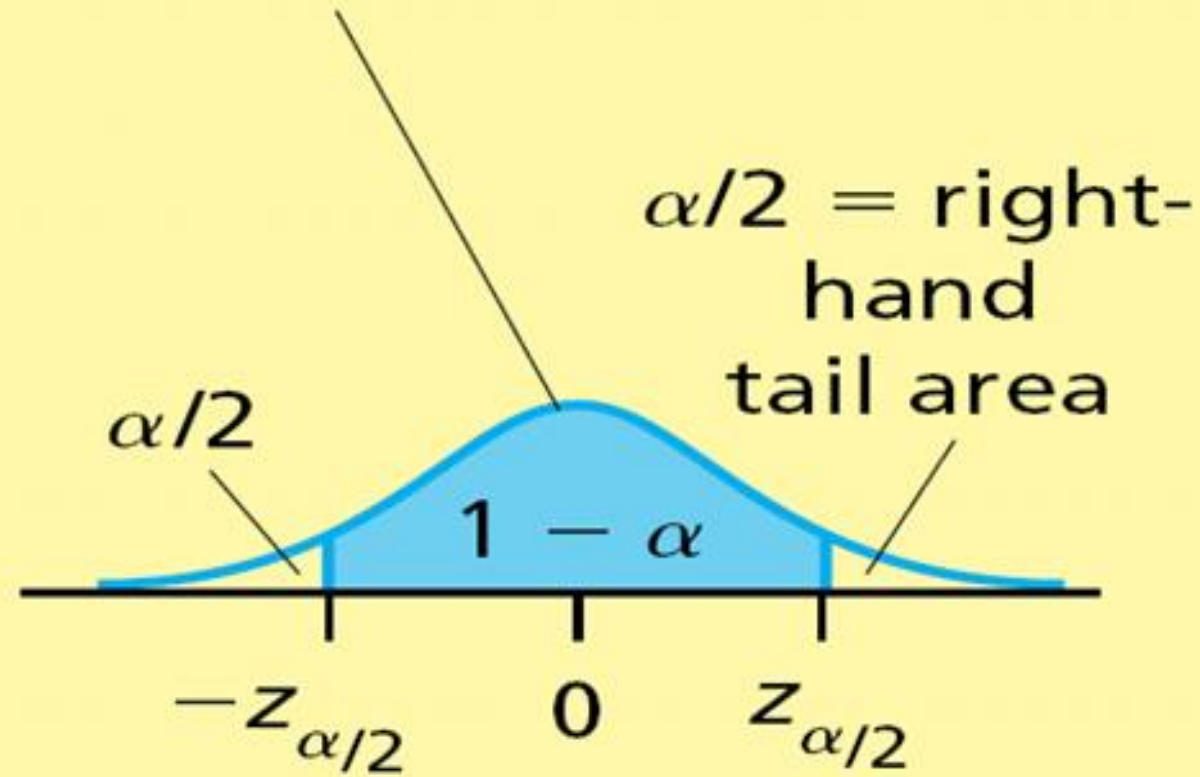
Recall General Confidence Interval we studied in Chapter 6 (e.g., Slide 51 of Chapter 6)

- In general, the probability is $1 - \alpha$ that the population mean μ is contained in the interval

$$[\bar{x} - z_{\alpha/2}\sigma_{\bar{x}}, \bar{x} + z_{\alpha/2}\sigma_{\bar{x}}] = \left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

- The normal point $z_{\alpha/2}$ gives a right-hand tail area under the standard normal curve equal to $\alpha/2$.
- The normal point $-z_{\alpha/2}$ gives a left-hand tail area under the standard normal curve equal to $\alpha/2$
- The area under the standard normal curve between $-z_{\alpha/2}$ and $z_{\alpha/2}$ is $1 - \alpha$
- $z_{\alpha/2}\sigma_{\bar{x}}$ is the **margin of error**.

Curve of the
standard normal
distribution



z-Based Confidence Intervals for a Mean with σ Known

- If a population has standard deviation σ (known),
- and if the population is normal or if sample size is large ($n \geq 30$), then ...
- ... a $100(1 - \alpha)$ percent **confidence interval for μ** is

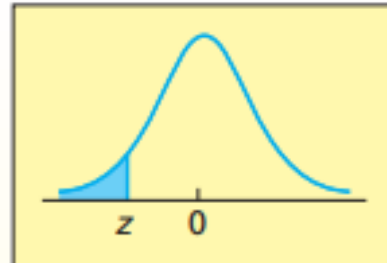
$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = \left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

For example, 95% Confidence Level

- For a 95% confidence level, $1 - \alpha = 0.95$, $\alpha = 0.05$, $\frac{\alpha}{2} = 0.025$
- For 95% confidence,
 - The area under the standard normal curve between $-z_{0.025}$ and $z_{0.025}$ is 0.95.
 - Then the area under the standard normal curve between negative infinite and $-z_{0.025}$ is 0.025.
 - Therefore, need the normal point $-z_{0.025}$.
 - Then $-z_{0.025} = -1.96$. Hence, $z_{0.025} = 1.96$.
 - The 95% confidence interval is

$$[\bar{x} \pm z_{0.025} \sigma_{\bar{x}}] = \left[\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \right] = \left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right]$$

TABLE A.3 Cumulative Areas under the Standard Normal Curve



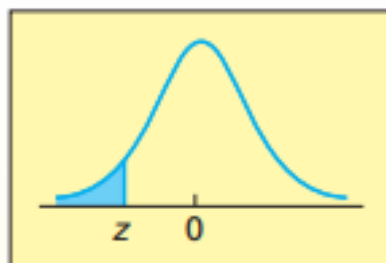
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
<u>-1.9</u>	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	<u>0.0250</u>	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294

For example, 99% Confidence Level

- For 99% confidence, need the normal point $z_{0.005}$
 - Reading between table entries in the standard normal table, the area is 0.005 for $-z_{0.005} = -2.575$
- The 99% confidence interval is

$$[\bar{x} \pm z_{0.005} \sigma_{\bar{x}}] = \left[\bar{x} \pm 2.575 \frac{\sigma}{\sqrt{n}} \right] = \left[\bar{x} - 2.575 \frac{\sigma}{\sqrt{n}}, \bar{x} + 2.575 \frac{\sigma}{\sqrt{n}} \right]$$

TABLE A.3 Cumulative Areas under the Standard Normal Curve



<i>z</i>	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
<u>-2.5</u>	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	<u>0.0051</u>	<u>0.0049</u>	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294

A general confidence interval procedure (for 95% confidence interval as an example)

Step 1: Place a symmetrical area of 0.95 under the standard normal curve and find the area in the normal curve tails beyond the 0.95 area. Because the entire area under the standard normal curve is 1, the area in both normal curve tails is $1 - 0.95 = 0.05$, and the area in each tail is 0.025.

Step 2: Find the normal point $z_{0.025}$ that gives a right-hand tail area under the standard normal curve equal to 0.025, and find the normal point $-z_{0.025}$ that gives a left-hand tail area under the curve equal to 0.025. The area under the standard normal curve between $z_{0.025}$ and $-z_{0.025}$ is 0.95, and the area under this curve to the left of $-z_{0.025}$ is 0.025. Looking up a cumulative area of 0.025 in Table A.3, we find that $-z_{0.025} = -1.96$.

Step 3: Form the following 95 percent confidence interval for the population mean

$$[\bar{x} \pm z_{0.025}\sigma_{\bar{x}}] = \left[\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \right]$$

If all possible samples were used to calculate this interval, then 95 percent of the resulting intervals would contain the population mean.

FIGURE 8.3 The Point $z_{.025}$

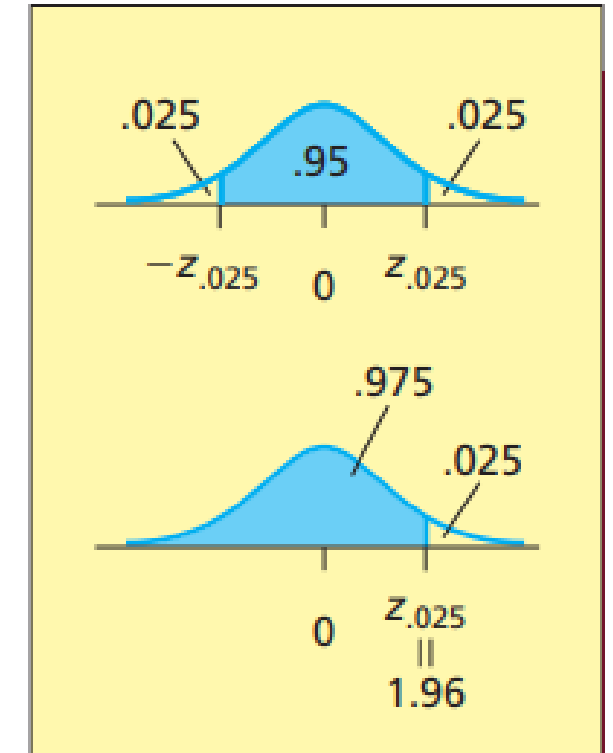
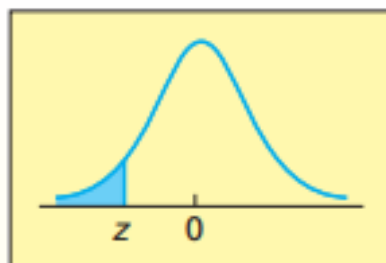


TABLE A.3 Cumulative Areas under the Standard Normal Curve



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-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
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Example 8.1 The e-billing Case: Reducing Mean Bill Payment Time

Recall that a management consulting firm has installed a new computer-based electronic billing system. The population mean payment time using the old billing system was approximately equal to, but no less than, 39 days. In order to assess whether the population mean payment time, μ , using the new billing system is substantially less than 39 days, the consulting firm will use the sample of $n = 65$ payment times in Table 2.4 to find a 99 percent confidence interval for μ . The mean of the 65 payment times is $\bar{x} = 18.1077$ days, and we will assume that the true value of the population standard deviation σ for the new billing system is 4.2 days (as discussed in Chapter 7). Then, because we previously showed that the normal point corresponding to 99 percent confidence is $z_{\alpha/2} = z_{0.005} = 2.575$, a 99 percent confidence interval for μ is

$$[\bar{x} \pm z_{0.005}\sigma_{\bar{x}}] = \left[18.1077 \pm 2.575 \frac{4.2}{\sqrt{65}} \right] = [16.8, 19.4]$$

So, we are 99 percent confident that the population mean payment time using the new billing system is between 16.8 days and 19.4 days. Therefore, we are 99 percent confident that the new billing system reduces the mean payment time by at most $39 - 16.8 = 22.2$ days and by at least $39 - 19.4 = 19.6$ days.

Example 8.1 The e-billing Case: Reducing Mean Bill Payment Time

To find a 95 percent confidence interval for μ .

$$[\bar{x} \pm z_{0.025}\sigma_{\bar{x}}] = \left[18.1077 \pm 1.96 \frac{4.2}{\sqrt{65}} \right] = [17.1, 19.1]$$

So, we are 95 percent confident that the population mean payment time using the new billing system is between 17.1 days and 19.1 days. Therefore, we are 95 percent confident that the new billing system reduces the mean payment time by at most $39 - 17.1 = 21.4$ days and by at least $39 - 19.4 = 19.9$ days.

It can be seen that the 99% confidence interval is slightly wider than the 95% confidence interval. The higher the confidence level, the wider the interval.

A Confidence Interval for a Population Mean μ : σ Known - Recap

Suppose that the sampled population is normally distributed with mean μ and standard deviation σ . Then a $100(1 - \alpha)$ percent confidence interval for μ is

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = \left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

Here, $z_{\alpha/2}$ is the normal point that gives a **right-hand tail area** under the standard normal curve. The normal point $z_{\alpha/2}$ can be found by looking up a cumulative area of $1 - \alpha/2$ in Table A.3. This confidence interval is also approximately valid for non-normal populations if the sample size is large (at least 30).

Note that the normal curve is symmetric. Hence, $-z_{\alpha/2}$ is the normal point that gives a **left-hand tail area** under the standard normal curve. The normal point $-z_{\alpha/2}$ can be found by looking up a cumulative area of $\alpha/2$ in Table A.3.

Exercises (if time permits)

8.1 Explain why it is important to calculate a confidence interval.

Answer: The sample mean $\bar{x}$ is not equal the true value of the population mean μ . By computing the confidence interval, we estimate — with confidence — the possible values that a population mean μ might equal.

Exercises (if time permits)

8.2 Explain the meaning of the term “95 percent confidence.”

Answer:

- We are 95% confident that the confidence interval contains the population mean, or
- The probability that the confidence interval contains the population mean is 95%.

Exercises (if time permits)

8.3 Under what conditions is the confidence interval $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ for μ valid?

Answer:

If the population is normally distributed, or
if the population is not normally distributed but n is bigger than 30

Exercises (if time permits)

8.4 For a fixed sample size, what happens to a confidence interval for μ when we increase the level of confidence?

Answer: The confidence interval is wider.

Exercises (if time permits)

8.5 For a fixed level of confidence, what happens to a confidence interval for μ when we increase the sample size?

Answer:

- As the sample size n increases, the standard deviation decreases.
- As the standard deviation decreases, the spread of the z curve decreases.
- As a result, the confidence interval for the population mean μ decreases.

Exercises (if time permits)

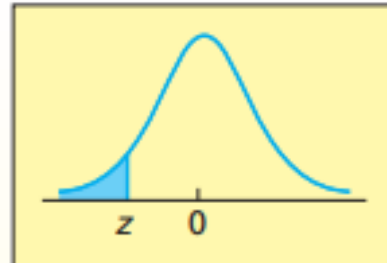
8.6 Suppose that, for a sample of size $n = 100$ measurements, we find that $\bar{x} = 50$. Assuming that σ equals 2, calculate confidence intervals for the population mean μ with the following confidence level 97%:

Answer:

- For 97% confidence, need the normal point $z_{0.015}$
 - Reading between table entries in the standard normal table, the area is 0.015 for $-z_{0.015} = -2.1$
- The 97% confidence interval is

$$[\bar{x} \pm z_{0.015} \sigma_{\bar{x}}] = \left[\bar{x} \pm 2.1 \frac{\sigma}{\sqrt{n}} \right] = \left[50 \pm 2.1 \frac{2}{\sqrt{100}} \right] = [49.58, 50.42]$$

TABLE A.3 Cumulative Areas under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
<u>-2.1</u>	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	<u>0.0150</u>	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294

Exercises (if time permits)

8.7 Consider the trash bag problem. The producer feels sure that its bags will be the strongest on the market if the new trash bag's mean breaking strength can be shown to be at least 50 pounds. The mean of the sample of 40 trash bag breaking strengths is $\bar{x} = 50.575$. If we let μ denote the mean of the breaking strengths of all possible trash bags of the new type and assume that σ equals 1.65:

- a** Calculate 95 percent and 99 percent confidence intervals for μ .
- b** Using the 95 percent confidence interval, can we be 95 percent confident that μ is at least 50 pounds? Explain.
- c** Using the 99 percent confidence interval, can we be 99 percent confident that μ is at least 50 pounds? Explain.
- d** Based on your answers to parts b and c, how convinced are you that the new bag is the strongest on the market?

Answer:

a For 95% confidence, need the normal point $z_{0.025}$

Reading between table entries in the standard normal table, the area is 0.025 for $-z_{0.025} = -1.96$

The 95% confidence interval is

$$[\bar{x} \pm z_{0.025} \sigma_{\bar{x}}] = \left[\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \right] = \left[50.575 \pm 1.96 \frac{1.65}{\sqrt{40}} \right] = [50.063, 51.086]$$

For 99% confidence, need the normal point $z_{0.005}$

Reading between table entries in the standard normal table, the area is 0.005 for $-z_{0.005} = -2.575$

The 99% confidence interval is

$$[\bar{x} \pm z_{0.005} \sigma_{\bar{x}}] = \left[\bar{x} \pm 2.575 \frac{\sigma}{\sqrt{n}} \right] = \left[50.575 \pm 2.575 \frac{1.65}{\sqrt{40}} \right] = [49.96, 51.19]$$

- b** Yes, all values in the interval $[50.063, 51.086]$ is higher than 50
- c** Yes, some values in the interval $[49.96, 51.19]$ is lower than 50
- b** Higher than 95% but smaller than 99% confident.

Challenge problem: For exercise 8.7, obtain the confidence level for which that μ is at least 50 pounds.

Step 1: Find $z_{\alpha/2}$

$$\begin{aligned} [\bar{x} \pm z_{\alpha/2} \sigma_{\bar{x}}] &= \left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] \\ &= \left[50.575 \pm z_{\alpha/2} \frac{1.65}{\sqrt{40}} \right] \end{aligned}$$

So that

$$50.575 - z_{\alpha/2} \frac{1.65}{\sqrt{40}} = 50$$

$$-z_{\alpha/2} = -2.204$$

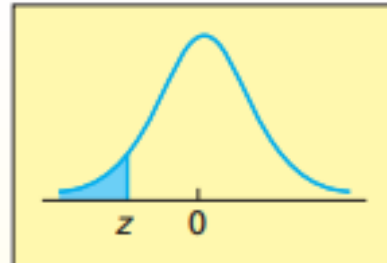
Step 2: Find the area of the standard normal distribution between $-z_{\alpha/2}$ and $z_{\alpha/2}$.

From the form,

- the area left-hand to -2.204 is 0.0125, the area right-hand to 2.204 is 0.0125
- So the area between -2.204 and 2.204 is $(1 - 0.0125 - 0.0125) = 0.975$

Hence, the confidence level is 97.5%.

TABLE A.3 Cumulative Areas under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
<u>-2.2</u>	0.0139	0.0136	0.0132	0.0129	<u>0.0125</u>	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294

In Section 8.1, we constructed **z**-Based Confidence Intervals for a Population Mean when **σ** is known

- It is based on the sampling distribution of

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- This sampling distribution is called the **z distribution**.

Section 8.2

t -Based Confidence Intervals for a Population Mean: σ unknown

- If σ is unknown (which is usually the case), we can construct a confidence interval for μ based on the sampling distribution of

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

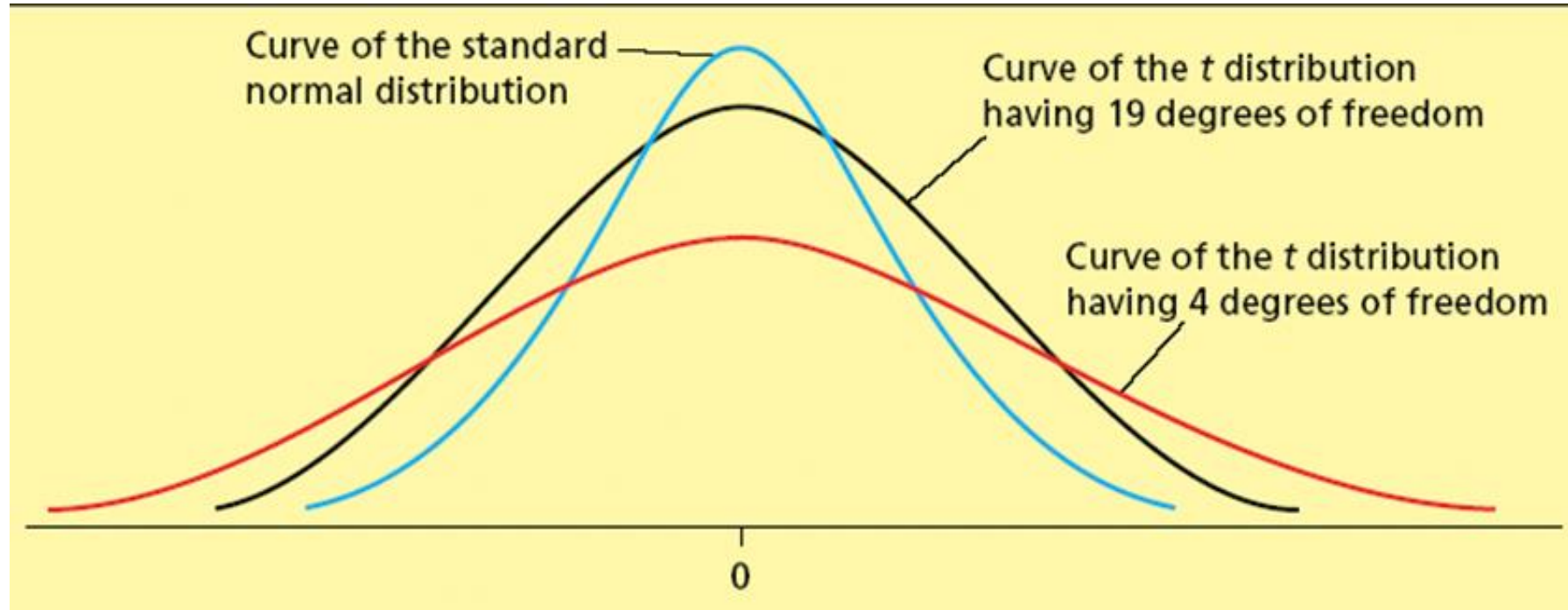
- If the population is normal, then for any sample size n , this sampling distribution is called the t distribution.

The t Distribution

- The curve of the t distribution is similar to that of the standard normal curve
 - Symmetrical and bell-shaped
 - The t distribution is more spread out than the standard normal distribution
 - The spread of the t is given by the **number of degrees of freedom**
 - Denoted by df
 - For a sample of size n , there are one fewer degrees of freedom, that is,

$$df = n - 1$$

Degrees of Freedom and the t-Distribution

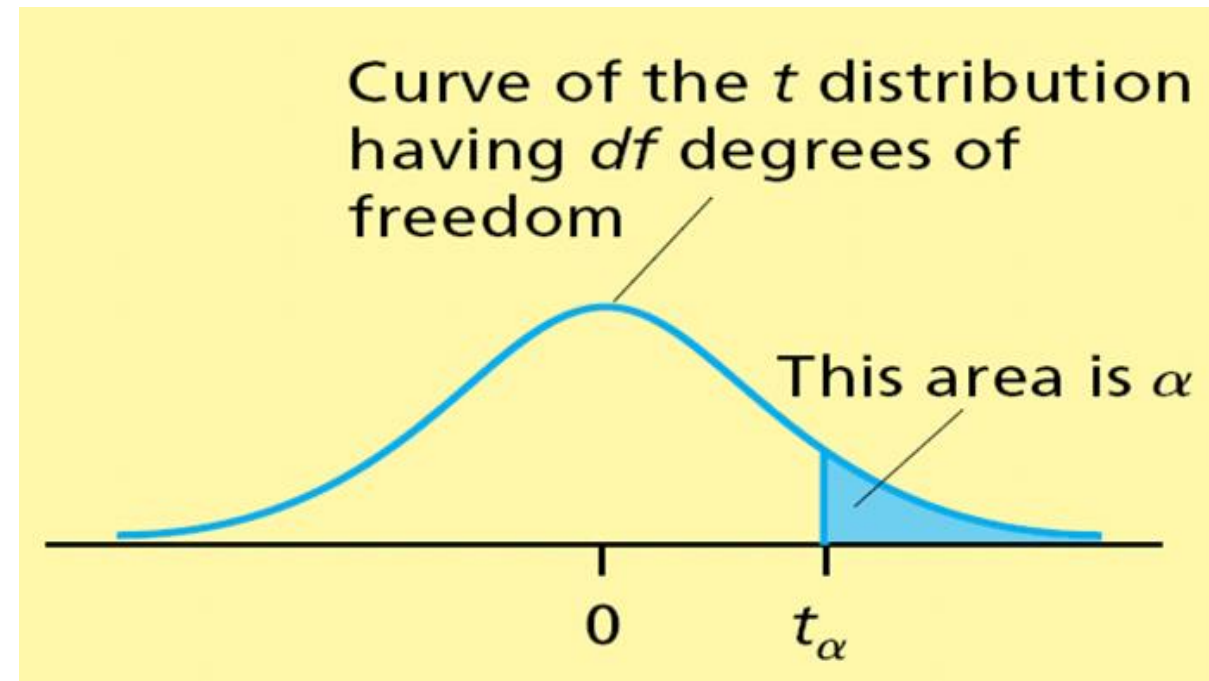


As the number of degrees of freedom increases, the spread of the t distribution decreases and the t curve approaches the standard normal curve.

- For a t distribution with $n - 1$ degrees of freedom,
 - As the sample size n increases, the degrees of freedom also increases
 - As the degrees of freedom increase, the spread of the t curve decreases
 - As the degrees of freedom increases indefinitely, the t curve approaches the standard normal curve
 - If $n \geq 30$, so $df = n - 1 \geq 29$, the t curve is very similar to the standard normal curve.

t and right-hand tail areas

- Use a t point denoted by t_α
 - t_α is the point on the horizontal axis under the t curve that gives a right hand tail equal to α
 - So the value of t_α in a particular situation depends on the right-hand tail area α and the number of degrees of freedom $df = n - 1$



Using the t Distribution Table

- Rows correspond to the different values of df
Columns correspond to different values of α

- See t Distribution Table in Appendix

The table gives t points for df 1 to 100, then for $df = 120$, and ∞

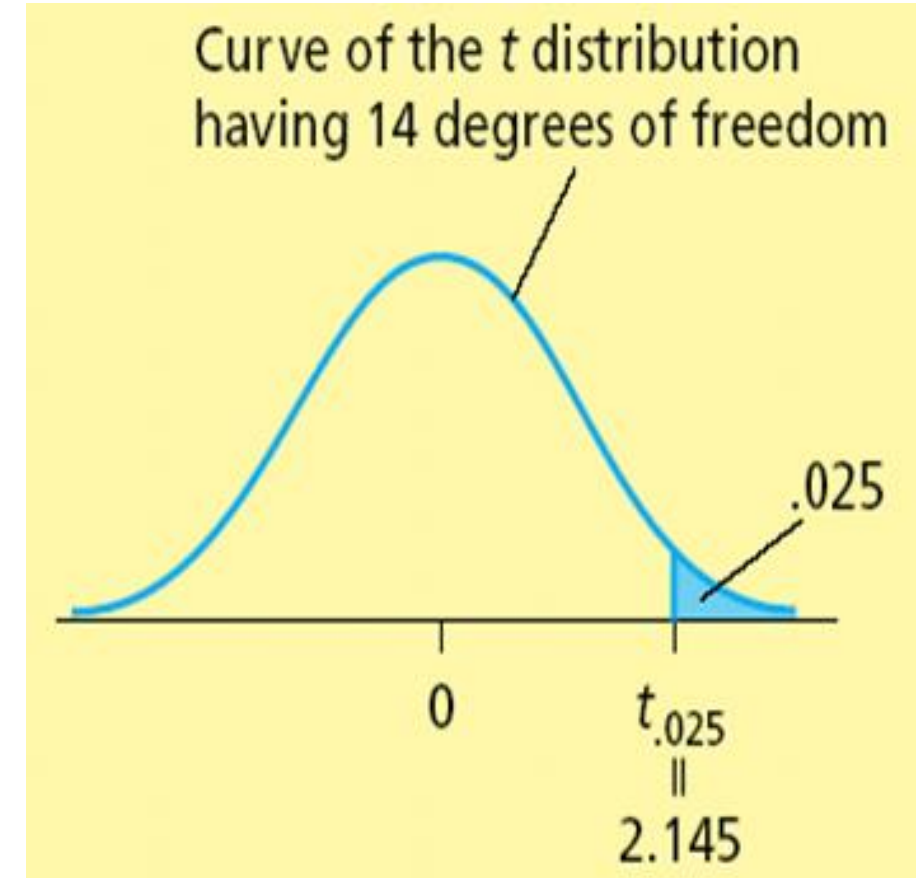
- On the row for ∞ , the t points are the z points
- Always look at the accompanying figure for guidance on how to use the table

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.01}$	$t_{.005}$	$t_{.001}$	$t_{.0005}$
1	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850

Example:

Find t_α for a sample of size $n = 15$ and right-hand tail area of 0.025

- For $n = 15$, $df = 14$
- $\alpha = 0.025$
 - Note that $\alpha = 0.025$ corresponds to a confidence level of 0.95
- In **t table**, along row labeled 14 and under column labeled 0.025, read a table entry of 2.145
- So $t_\alpha = 2.145$

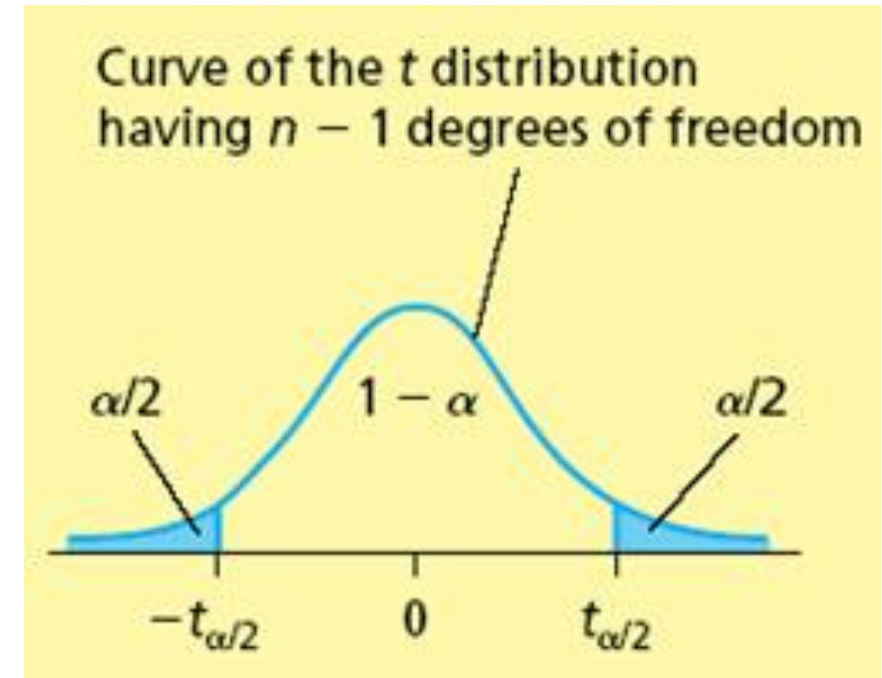


***t*-Based Confidence Intervals for a Mean: σ Unknown**

If the sampled population is normally distributed with mean μ , then a $100(1 - \alpha)$ percent confidence interval for μ is

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$t_{\alpha/2}$ is the t point giving a right-hand tail area of $\alpha/2$ under the t curve having $n - 1$ degrees of freedom



Example 8.4

Debt-to-Equity Ratio

One measure of a company's financial health is its *debt-to-equity ratio*. This quantity is defined to be the ratio of the company's corporate debt to the company's equity. If this ratio is too high, it is one indication of financial instability. For obvious reasons, banks often monitor the financial health of companies to which they have extended commercial loans. Suppose that, in order to reduce risk, a large bank has decided to initiate a policy limiting the mean debt-to-equity ratio for its portfolio of commercial loans to being less than 1.5. In order to estimate the mean debt-to-equity ratio of its (current) commercial loan portfolio, the bank randomly selects a sample of 15 of its commercial loan accounts. Audits of these companies result in the following debt-to-equity ratios:

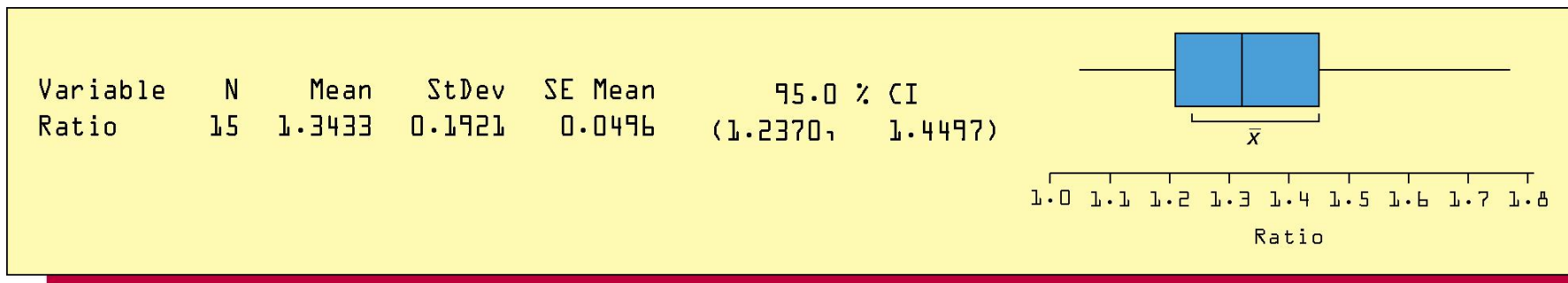
1.31	1.05	1.45	1.21	1.19
1.78	1.37	1.41	1.22	1.11
1.46	1.33	1.29	1.32	1.65

DS DebtEq

Example 8.4

Debt-to-Equity Ratio

- Estimate the mean debt-to-equity ratio of the loan portfolio of a bank
- Select a random sample of 15 commercial loan accounts
 - Box plot is given in figure below
- Know: $\bar{x} = 1.34$ $s = 0.192$ $n = 15$
- Want a 95% confidence interval for the ratio
- Assume all ratios are **normally** distributed but **σ unknown**



- Have to use the t distribution
- At 95% confidence,
 - $1 - \alpha = 0.95$ so $\alpha = 0.05$ and $\alpha/2 = 0.025$
- For $n = 15$,
 - $df = 15 - 1 = 14$
- Use the t table to find $t_{\alpha/2}$ for $df = 14$
 - $t_{\alpha/2} = t_{0.025} = 2.145$ for $df = 14$
- The 95% confidence interval:

$$\begin{aligned}
 \bar{x} \pm t_{0.025} \frac{s}{\sqrt{n}} &= 1.343 \pm 2.145 \frac{0.192}{\sqrt{15}} \\
 &= 1.343 \pm 0.106 \\
 &= [1.237, 1.449]
 \end{aligned}$$

This interval says the bank is 95 percent confident that the mean debt-to-equity ratio for its portfolio of commercial loan accounts is between 1.2369 and 1.4497. Based on this interval, the bank has strong evidence that the portfolio's mean ratio is less than 1.5 (or that the bank is in compliance with its new policy).

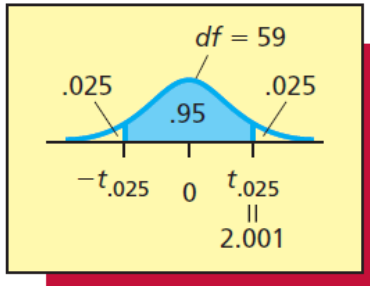
A green circle with a yellow border containing the white letters "BI".A green circle with a yellow border containing the white letters "BI".

(page 792). This interval says we are 95 percent confident that the model's population mean mileage μ is between 31.33 mpg and 31.79 mpg. Based on this interval, the model's EPA mileage estimate is 31 mpg, and the automaker will receive the tax credit.

A green circle with a yellow border containing the white letters "BI".

(page 792). Recalling that the mean payment time using the old billing system is 39 days, the interval says that we are 99 percent confident that the population mean payment time using the new billing system is between 16.8 days and 19.4 days. Therefore, we are 99 percent confident that the new billing system reduces the mean payment time by at most 22.2 days and by at least 19.6 days.

Example 8.5 The Marketing Research Case



Recall that a brand group is considering a new bottle design for a popular soft drink and that Table 1.5 (page 10) gives a random sample of $n = 60$ consumer ratings of this new bottle design. Let μ denote the mean rating of the new bottle design that would be given by all consumers. In order to assess whether μ exceeds the minimum standard composite score of 25 for a successful bottle design, the brand group will calculate a 95 percent confidence interval for μ . The mean and the standard deviation of the 60 bottle design ratings are $\bar{x} = 30.35$ and $s = 3.1073$. It follows that a 95 percent confidence interval for μ is

$$\left[\bar{x} \pm t_{.025} \frac{s}{\sqrt{n}} \right] = \left[30.35 \pm 2.001 \frac{3.1073}{\sqrt{60}} \right] = [29.5, 31.2]$$

where $t_{.025} = 2.001$ is based on $n - 1 = 60 - 1 = 59$ degrees of freedom—see Table A.4 (page 792). Because the interval says we are 95 percent confident that the population mean rating of the new bottle design is between 29.5 and 31.2, we are 95 percent confident that this mean rating exceeds the minimum standard of 25 by at least 4.5 points and by at most 6.2 points.



8.3 Sample Size Determination

In Section 8.1 we used a sample of 50 mileages to construct a 95 percent confidence interval for the midsize model's mean mileage μ . The size of this sample was not arbitrary—it was

30 mpg, and the automaker would not receive its tax credit. One reason that the lower limit of this 95 percent interval is less than 31 is that the sample size of 5 is not large enough to make the interval's margin of error

$$z_{.025} \frac{\sigma}{\sqrt{n}} = 1.96 \frac{.8}{\sqrt{5}} = .701$$

small enough. We can attempt to make the margin of error in the interval smaller by increasing the sample size. If we feel that the mean $\bar{x}$ of the larger sample will be at least 31.3 mpg (the mean of the small sample we have already taken), then the lower limit of a $100(1 - \alpha)$ percent confidence interval for μ will be at least 31 if the margin of error is .3 or less.

for μ is $[\bar{x} \pm z_{\alpha/2} (\sigma / \sqrt{n})]$. To find the needed sample size, we set $z_{\alpha/2} (\sigma / \sqrt{n})$ equal to the desired margin of error and solve for n . Letting E denote the desired margin of error, we obtain

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = E$$

Multiplying both sides of this equation by $\sqrt{n}$ and dividing both sides by E , we obtain

$$\sqrt{n} = \frac{z_{\alpha/2} \sigma}{E}$$

Squaring both sides of this result gives us the formula for n .

8.3 Sample Size Determination

Determining the Sample Size for a Confidence Interval for μ : σ Known

A sample of size

$$n = \left(\frac{Z_{\alpha/2}\sigma}{E} \right)^2$$

makes the margin of error in a $100(1 - \alpha)$ percent confidence interval for μ equal to E . That is, this sample size makes us $100(1 - \alpha)$ percent confident that $\bar{x}$ is within E units of μ . If the calculated value of n is not a whole number, round this value up to the next whole number (so that the margin of error is at least as small as desired).

If σ is unknown and is estimated from s , then a sample of size

$$n = \left(\frac{t_{\alpha/2} s}{E} \right)^2$$

so that $\bar{x}$ is within E units of μ , with $100(1-\alpha)\%$ confidence. The number of degrees of freedom for the $t_{\alpha/2}$ point is the size of the preliminary sample minus 1

Example 8.6

Car Mileage Case

Suppose in the car mileage situation we wish to find the sample size that is needed to make the **margin of error** in a **95 percent** confidence interval for μ equal to **0.3**. Assuming that **σ is known to equal 0.8**.

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 = \left(\frac{z_{0.025} \sigma}{E} \right)^2 = \left(\frac{1.96(0.8)}{0.3} \right)^2 = 27.32$$

Rounding up, we would employ a sample of size 28

Car Mileage Case

- Suppose in the car mileage situation we wish to find the sample size that is needed to make the **margin of error** in a **95 percent** confidence interval for μ equal to **0.3**. Assuming that **σ is unknown**.
- We regard the sample of five mileages as a preliminary sample with standard deviation $s=0.7583$.

$$n = \left(\frac{t_{\alpha/2} s}{E} \right)^2 = \left(\frac{2.776(0.7583)}{0.3} \right)^2 = 49.24$$

Rounding up, we employ a sample of size 50.

95 percent of all possible sample means based on 50 mileages are within 0.3 of μ

Finally, sometimes we do not know σ and we do not have a preliminary sample that can be used to estimate σ . In this case it can be shown that, if we can make a reasonable guess of the range of the population being studied, then a conservatively large estimate of σ is this estimated range divided by 4. For example, if the automaker's design engineers feel that almost all of its midsize cars should get mileages within a range of 5 mpg, then a conservatively large estimate of σ is $5/4 = 1.25$ mpg. When employing such an estimate of σ , it is sufficient to use the z -based sample size formula $n = (z_{\alpha/2}\sigma/E)^2$, because a conservatively large estimate of σ will give us a conservatively large sample size.

8.4 Confidence Intervals for a Population Proportion

In order to see if there is strong evidence that p is less than .10, we can calculate a confidence **interval for p** . As explained in Chapter 7, if the sample size n is large, then the sampling distribution of the sample proportion **$\hat{p}$ is approximately a normal distribution** with mean $\mu_{\hat{p}} = p$ and standard deviation $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$. By using the same logic we used in de-

If the sample size n is large*, then a **$(1-\alpha)100\%$ confidence interval for p** is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

* Here n should be considered large if both $n \cdot \hat{p} \geq 5$ and $n \cdot (1 - \hat{p}) \geq 5$

Example 8.8: The Cheese Spread Case

$$\begin{aligned}\left[\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] &= \left[.063 \pm 2.576 \sqrt{\frac{(.063)(.937)}{1,000}} \right] \\ &= [.063 \pm .0198] \\ &= [.0432, .0828]\end{aligned}$$

all current purchasers would stop buying the cheese spread if the new spout were used. Moreover, because the upper limit of the 99 percent confidence interval is less than .10, we have very strong evidence that the true proportion p of all current purchasers who would stop buying the cheese spread is less than .10. Based on this result, it seems reasonable to use the new spout.



The Phe-Mycin Case: Drug Side Effects

Antibiotics occasionally cause nausea as a side effect. Scientists working for a major drug company have developed a new antibiotic called Phe-Mycin. The company wishes to estimate p , the proportion of all patients who would experience nausea as a side effect when being treated with Phe-Mycin. Suppose that a sample of 200 patients is randomly selected. When these patients are treated with Phe-Mycin, 35 patients experience nausea. The point estimate of the population proportion p is the sample proportion $\hat{p} = 35/200 = .175$. This sample proportion says that we estimate that 17.5 percent of all patients would experience nausea as a side effect of taking Phe-Mycin. Furthermore, because $n\hat{p} = 200(.175) = 35$ and $n(1 - \hat{p}) = 200(.825) = 165$ are

Phe-Mycin Side Effects

Given: $n = 200$, 35 patients experience nausea.

$$\hat{p} = \frac{35}{200} = 0.175$$

Note:

$$n\hat{p} = 200 \times 0.175 = 35$$

$$n(1 - \hat{p}) = 200 \times 0.825 = 165$$

so both quantities are > 5

For 95% confidence, $z_{\alpha/2} = z_{0.025} = 1.96$ and

$$\begin{aligned}\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} &= 0.175 \pm 1.96 \sqrt{\frac{0.175 \times 0.825}{200}} \\ &= 0.175 \pm 0.053 \\ &= [0.122, 0.228]\end{aligned}$$

Example 8.10

The Marketing Ethics Case: Confidentiality



In the book *Essentials of Marketing Research*, William R. Dillon, Thomas J. Madden, and Neil H. Firtle discuss a survey of marketing professionals, the results of which were originally published by Ishmael P. Akoah and Edward A. Riordan in the *Journal of Marketing Research*. In the study, randomly selected marketing researchers were presented with various scenarios involving ethical issues such as confidentiality, conflict of interest, and social acceptability. The marketing researchers were asked to indicate whether they approved or disapproved of the actions described in each scenario. For instance, one scenario that involved the issue of confidentiality was described as follows:

Use of ultraviolet ink A project director went to the marketing research director's office and requested permission to use an ultraviolet ink to precode a questionnaire for a mail survey. The project director pointed out that although the cover letter promised confidentiality, respondent identification was needed to permit adequate cross-tabulations of the data. The marketing research director gave approval.

Of the 205 marketing researchers who participated in the survey, 117 said they disapproved of the actions taken in the scenario. It follows that a point estimate of p , the proportion of all marketing researchers who disapprove of the actions taken in the scenario, is $\hat{p} = 117/205 = .5707$. Furthermore, because $n\hat{p} = 205(.5707) = 117$ and $n(1 - \hat{p}) = 205(.4293) = 88$ are both at least 5, a 95 percent confidence interval for p is

$$\begin{aligned} \left[\hat{p} \pm z_{.025} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right] &= \left[.5707 \pm 1.96 \sqrt{\frac{(.5707)(.4293)}{205}} \right] \\ &= [.5707 \pm .0678] \\ &= [.5029, .6385] \end{aligned}$$

This interval says we are 95 percent confident that between 50.29 percent and 63.85 percent of all marketing researchers disapprove of the actions taken in the ultraviolet ink scenario. Notice that because the margin of error (.0678) in this interval is rather large, this interval does not provide a very precise estimate of p . Below we show the MINITAB output of this interval.

CI for One Proportion			
X	N	Sample p	95% CI
117	205	0.570732	(0.502975, 0.638488)

Determining the Sample Size for a Confidence Interval for p

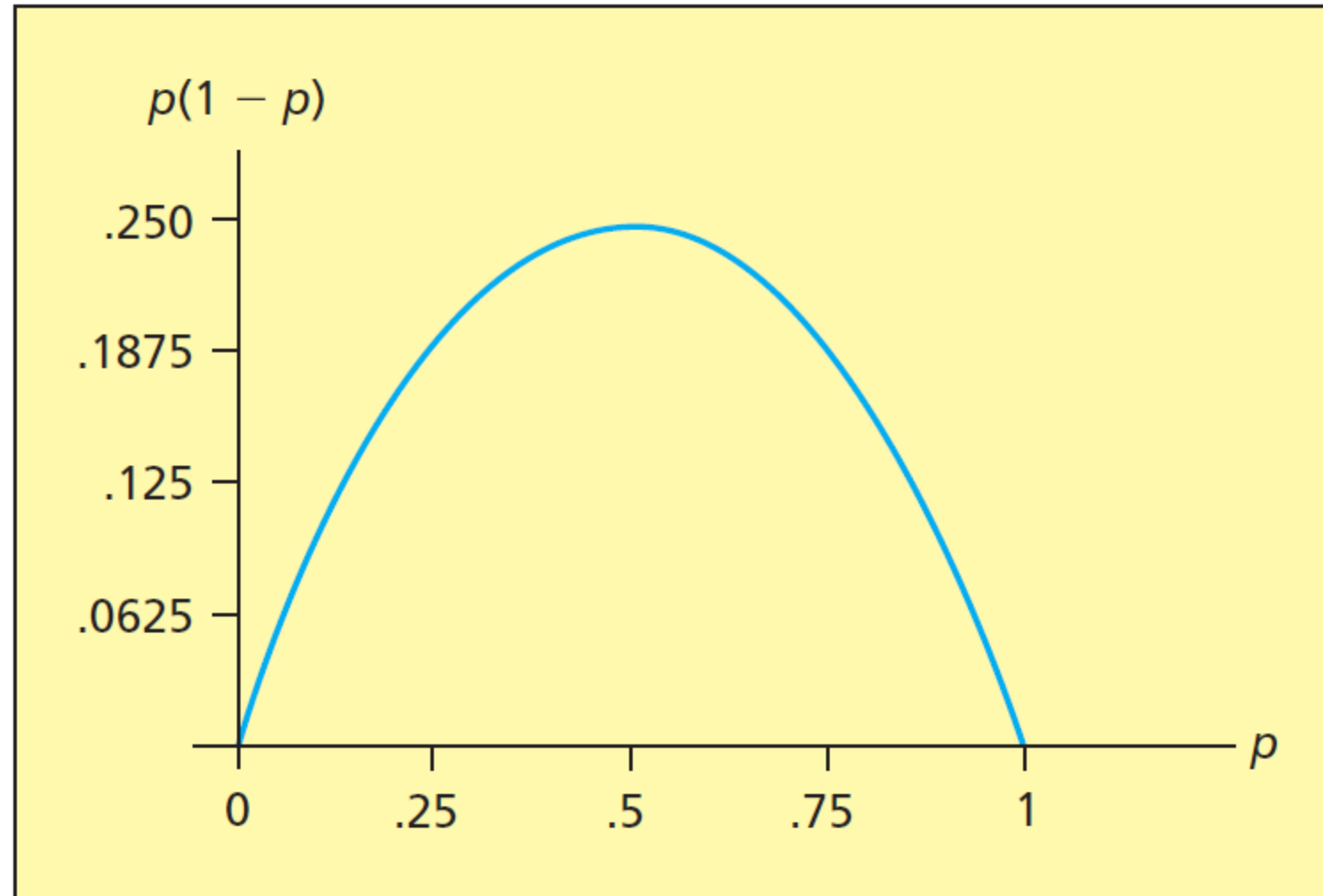
A sample of size

$$n = p(1 - p) \left(\frac{z_{\alpha/2}}{E} \right)^2$$

makes the margin of error in a $100(1 - \alpha)$ percent confidence interval for p equal to E . That is, this sample size makes us $100(1 - \alpha)$ percent confident that $\hat{p}$ is within E units of p . If the calculated value of n is not a whole number, round this value up to the next whole number.

Note that the formula requires a preliminary estimate of p . The conservative value of $p = 0.5$ is generally used when there is no prior information on p .

FIGURE 8.16 The Graph of $p(1 - p)$ versus p

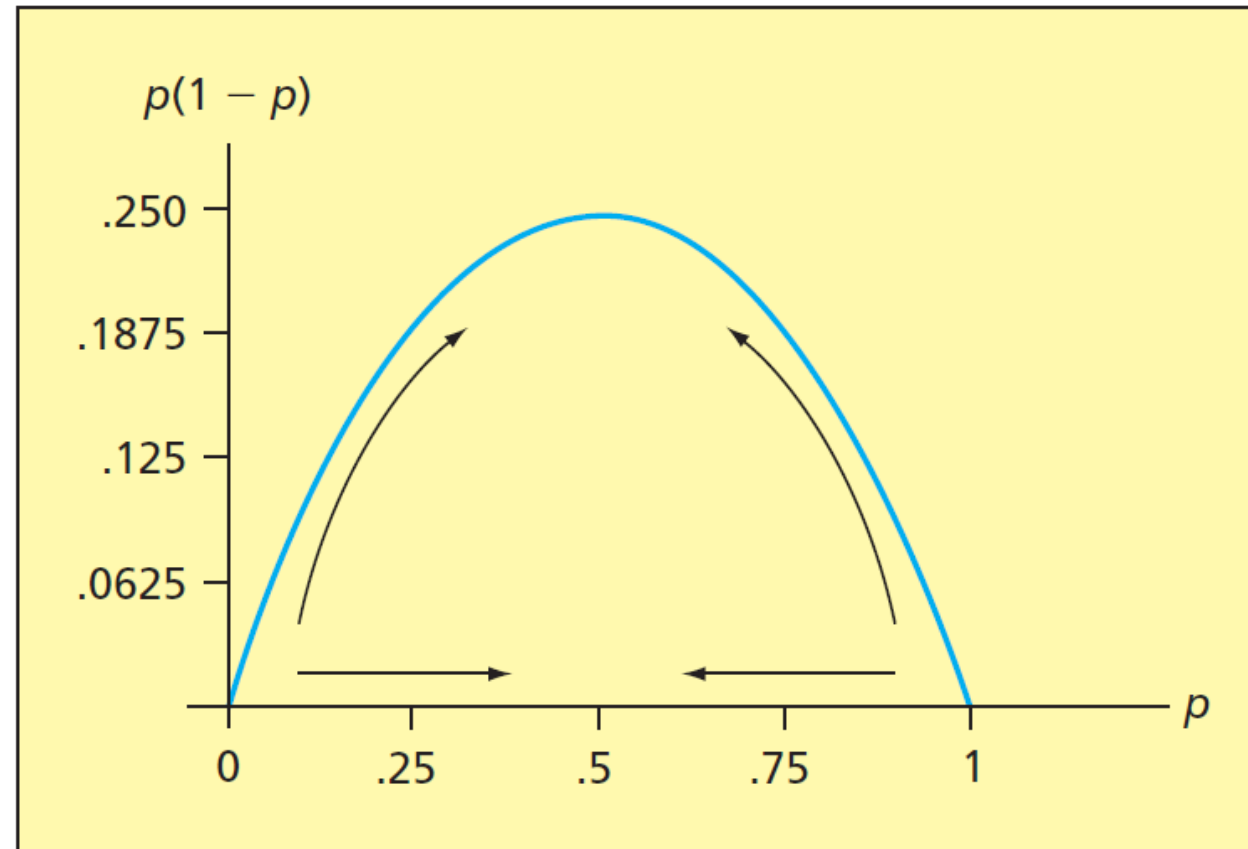


could be near .5. Furthermore, suppose we wish to make the margin of error in a 95 percent confidence interval for p equal to .02. If the sample to be taken is random, it should consist of

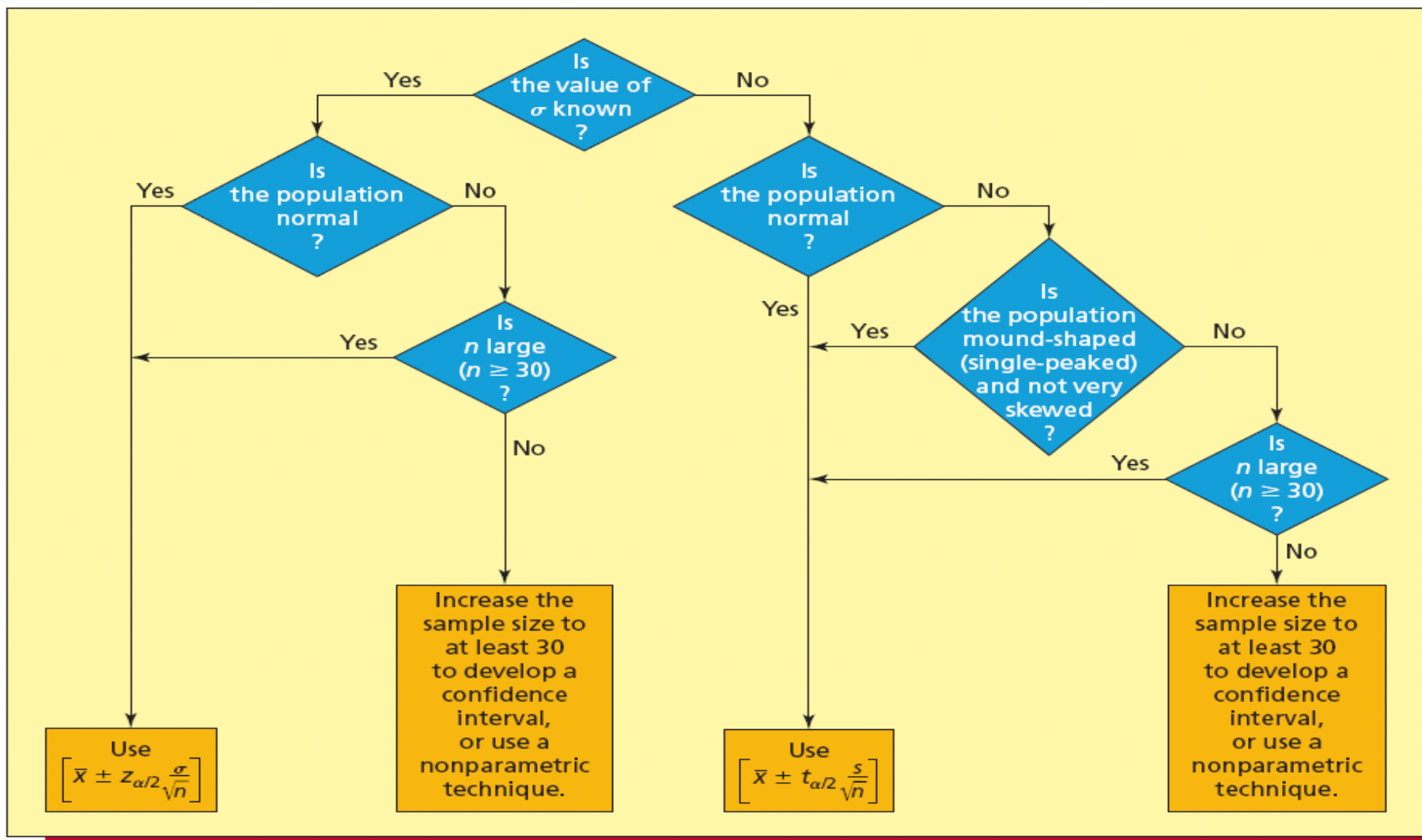
$$n = p(1 - p) \left(\frac{z_{\alpha/2}}{E} \right)^2 = .25 \left(\frac{1.96}{.02} \right)^2 = 2,401$$

If the population proportion we are estimating is substantially different from .5, setting p equal to .5 will give a sample size that is much larger than is needed. In this case, we should use our intuition or previous sample information (along with Figure 8.17) to determine the largest reasonable value for $p(1 - p)$. Figure 8.17 implies that as p gets closer to .5, $p(1 - p)$ increases. It follows that $p(1 - p)$ is maximized by the reasonable value of p that is closest to .5. Therefore, **when we are estimating a proportion that is substantially different from .5, we use the reasonable value of p that is closest to .5 to calculate the sample size needed to obtain a specified margin of error.**

FIGURE 8.17 As p Gets Closer to .5, $p(1 - p)$ Increases



Summary: Selecting an Appropriate Confidence Interval for a Population Mean



Example: Suppose an insurance company studies repair costs after rear collisions, and finds the mean repair cost to be \$2300 based on a sample of 40 accidents. Suppose the standard deviation of population is \$1025.
Find the 95% Confidence Interval.

A sample of 500 executives who own their own home revealed 175 planned to sell their homes and retire to Arizona. Develop a 98% confidence interval for the proportion of executives that plan to sell and move to Arizona.



The American Kennel Club wanted to estimate the proportion of children that have a dog as a pet. If the club wanted the estimate to be within 3% of the population proportion, how many children would they need to contact?



Assume a 95% level of confidence and that the club estimated that 30% of the children have a dog as a pet.

Chapter Summary

In this chapter we discussed **confidence intervals** for population **means** and **proportions**. First, we studied how to compute a confidence interval for a **population mean**. We saw that when the population standard deviation σ is known, we can use the **normal distribution** to compute a confidence interval for a population mean. When σ is not known, if the population is normally distributed (or at least mound-shaped) or if the sample size n is large, we use the **t distribution** to compute this interval. We also studied how to find the size of the sample needed if we wish to compute a confidence interval for a mean with a pre-specified *confidence level* and with a prespecified *margin of*

error. Figure 8.18 is a flowchart summarizing our discussions concerning how to compute an appropriate confidence interval for a population mean.

Next we saw that we are often interested in estimating the proportion of population units falling into a category of interest. We showed how to compute a large sample confidence interval for a **population proportion**, and we saw how to find the sample size needed to estimate a population proportion. We concluded this chapter with an optional section that discusses how to estimate means, proportions, and totals for finite populations.

Thank you!